Project: Worked examples
Title: Load budget, PSU, transformer and battery
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

The demand of a set of loads per supply (three-phase AC, single-phase AC, DC) and the minimum DC power supply, transformer and battery for it, each with a margin; the heat the loads dissipate for the enclosure calculation.
Demand per load: P_i·d_i·k_i; apparent power P_i·d_i·k_i/cosφ_i (arithmetic sum, conservative) Bus currents: I_3φ = S_3φ/(√3·U), I_1φ = S_1φ/U, I_DC = P_DC/U_DC; when a bus is chosen for the DC power supply its input P_DC/η_PSU (power factor 1) is added to that bus Heat dissipated inside the enclosure: Φ = Σ h_i·P_i·d_i·k_i, h_i = the share of the input power released inside (blank: the loss 1 − η_i; electronics 1; a machine outside the enclosure 0) PSU: P_PSU ≥ margin·P_DC/f_derate (f_derate from the manufacturer's curve at θ and altitude) Transformer: S_T ≥ margin·S for the running load; starting peak S_peak = S + max(m_i·P_i/cosφ_i − S_i) ≤ k_T·S_T (the starting load undiversified), so S_T ≥ S_peak/k_T for the start; the rating to buy is the larger of the two Battery: C = I_b·t/(DoD·η_b·k_temp·k_age), I_b = Σ backed P_i·d_i·k_i/(U_b·η_conv); Peukert: I_b·H·(t/H)^(1/k_p) in place of I_b·t
Ref: Linden's Handbook of Batteries 5th ed., Ch. 3–4 — Battery sizing and the Peukert relation
Ref: BS EN 60204-1:2018 cl. 4.3 — Electrical supply (supply voltage tolerance; cited by number only)

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
loads	Loads	6 row(s)		drawing: E-SCH-010 rev A	any
loads[1]		name = Duty pump; supply = AC three-phase; P = 11 kW; cosphi = 0.86; eta = 0.91; d = 1; k = 1; inrush = 6.5; backed = no			

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
loads[2]		name = Standby pump; supply = AC three-phase; P = 11 kW; cosphi = 0.86; eta = 0.91; d = 1; k = 0; inrush = 6.5; backed = no			
loads[3]		name = Panel fan; supply = AC single-phase; P = 0.06 kW; cosphi = 0.9; eta = 0.7; d = 1; k = 1; inrush = 1; backed = no			
loads[4]		name = Panel light; supply = AC single-phase; P = 0.02 kW; cosphi = 0.95; eta = 1; d = 0.1; k = 1; inrush = 1; backed = no			

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
loads[5]		name = PLC and I/O; supply = DC; P = 0.096 kW; cosphi = 1; eta = 0.9; d = 1; k = 1; inrush = 1; backed = yes			
loads[6]		name = Router; supply = DC; P = 0.024 kW; cosphi = 1; eta = 0.9; d = 1; k = 1; inrush = 1; backed = yes			
P_add	Additional three-phase load (linked from E1's RMS power)	0	kW	default	0 … 200 kW
cosphi_add	Power factor of the additional load	0.85		default	0.5 … 1
eta_add	Efficiency of the additional load	0.9		default	0.5 … 1
inrush_add	Starting multiple of the additional load	1		default	1 … 20
heat_add	Share of the additional load's input power released inside the enclosure (a linked machine is outside it: 0)	0		default	0 … 1
U_3ph	Three-phase line voltage	400	V	spec: Site supply	12 … 1000 V
U_1ph	Single-phase voltage	230	V	spec: Site supply	12 … 1000 V
U_DC	DC bus voltage	24	V	drawing: E-SCH-010 rev A	12 … 1000 V
psu_bus	AC bus that feeds the DC power supply (its input P_DC/η_PSU is added to that bus)	AC single-phase		drawing: E-SCH-010 rev A	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
eta_psu	DC power supply efficiency (power factor taken as 1)	0.9		assumption: 24 V supply efficiency 90 %, power factor taken as 1.	0.5 … 1
margin	Sizing margin	1.25		assumption: 25 % growth margin.	1 … 1.5
theta	Ambient temperature	45	degC	spec: Plant room design maximum	-20 … 70 degC
f_derate	PSU derating factor at the ambient and altitude	1		assumption: PSU rated to 60 °C without derating (datasheet).	0.1 … 1
P_PSU	Selected PSU rating	240	W	drawing: E-SCH-010 rev A	1 … 100000 W
S_T	Selected transformer rating	25	kVA	spec: Existing site transformer allocation	0.01 … 10000 kVA
k_T	Transformer short-time overload allowed for inrush	3		assumption: Short-time overload allowed by the supply authority.	1 … 20
t_auto	Battery autonomy	2	h	spec: Telemetry to stay up for 2 h on mains loss	0 … 72 h
U_b	Battery nominal voltage	24	V	drawing: E-SCH-010 rev A	2 … 1000 V
eta_conv	Conversion efficiency battery → loads	1		assumption: DC UPS module, loads on the battery bus.	0.5 … 1
DoD	Allowed depth of discharge	0.8		assumption: VRLA, 80 % depth of discharge.	0.1 … 1
eta_b	Battery discharge efficiency	0.95		assumption: Discharge efficiency.	0.5 … 1
k_temp	Temperature capacity factor	1		assumption: Battery kept at 20 °C.	0.3 … 1.2
k_age	Ageing (end-of-life) factor	0.8		assumption: End-of-life capacity 80 %.	0.5 … 1
C_batt	Selected battery capacity	17	A*h	drawing: E-SCH-010 rev A	0.1 … 100000 A*h

3. Assumptions

A1. Each load's P is its electrical input; DC loads count at power factor 1 in the transformer demand; with a

PSU bus chosen they count at P/η_PSU, the PSU's own loss being a load row if its heat is wanted. (Load budget convention.)
A2. The heat released inside the enclosure is h·P·d·k per load; the default h = 1 − η is the loss of a drive or supply, electronics release all their input (h = 1) and a machine outside the panel releases none (h = 0). (Stated convention for the heat a load releases inside the panel; change h for a load that differs.)
A3. Apparent powers are added arithmetically, which is conservative when power factors differ. (Upper bound on S.)
A4. PSU derating, transformer overload and battery factors are the manufacturer's values, entered as records. (Entered as reference records, which carry their source, edition and licence.)
A5. eta_psu (DC power supply efficiency (power factor taken as 1)) is taken as 0.9. (24 V supply efficiency 90 %, power factor taken as 1.)
A6. margin (Sizing margin) is taken as 1.25. (25 % growth margin.)
A7. f_derate (PSU derating factor at the ambient and altitude) is taken as 1. (PSU rated to 60 °C without derating (datasheet).)
A8. k_T (Transformer short-time overload allowed for inrush) is taken as 3. (Short-time overload allowed by the supply authority.)
A9. eta_conv (Conversion efficiency battery → loads) is taken as 1. (DC UPS module, loads on the battery bus.)
A10. DoD (Allowed depth of discharge) is taken as 0.8. (VRLA, 80 % depth of discharge.)
A11. eta_b (Battery discharge efficiency) is taken as 0.95. (Discharge efficiency.)
A12. k_temp (Temperature capacity factor) is taken as 1. (Battery kept at 20 °C.)
A13. k_age (Ageing (end-of-life) factor) is taken as 0.8. (End-of-life capacity 80 %.)

4. Working

[22] L1 = P·d·k; P·d·k/cos□
= 11 × 1 × 1 = 11 kW; 12.791 kVA
= 11 kW
Duty pump (AC three-phase)

[23] L2 = P·d·k; P·d·k/cos□
= 11 × 1 × 0 = 0 kW; 0 kVA
= 0 kW
Standby pump (AC three-phase)

[24] L3 = P·d·k; P·d·k/cos□
= 0.06 × 1 × 1 = 0.06 kW; 0.066667 kVA
= 0.06 kW
Panel fan (AC single-phase)

[25] L4 = P·d·k; P·d·k/cos□
= 0.02 × 0.1 × 1 = 0.002 kW; 0.0021053 kVA
= 0.002 kW
Panel light (AC single-phase)

[26] L5 = P·d·k; P·d·k/cos□
= 0.096 × 1 × 1 = 0.096 kW; 0.10667 kVA
= 0.096 kW
PLC and I/O (DC)

[27] L6 = P·d·k; P·d·k/cos□
= 0.024 × 1 × 1 = 0.024 kW; 0.026667 kVA
= 0.024 kW
Router (DC)

[28] P_conn = □ P_i
= 11 + 11 + 0.06 + 0.02 + 0.096 + 0.024
= 22.2 kW
Connected load

[29] P_dem = ∑ P_i·d_i·k_i
= 11 + 0 + 0.06 + 0.002 + 0.096 + 0.024
= 11.182 kW
Demand active power
[30] S_dem = ∑ P_i·d_i·k_i/cos∅_i
= 12.791 + 0 + 0.066667 + 0.0021053 + 0.10667 + 0.026667
= 12.993 kVA
Demand apparent power (arithmetic)
[31] I_3ph = S_3∅/(√3·U_3ph)
= 12791 / (√3 × 400)
= 18.462 A
Three-phase bus current
[32] I_1ph = S_1∅/U_1ph
= 202.11 / 230
= 0.87872 A
Single-phase bus current
[33] I_DC = P_DC/U_DC
= 120 / 24
= 5 A
DC bus current
[34] Phi = ∑ h_i·P_i·d_i·k_i
= 0.09×11 + 0.09×0 + 0.3×0.06 + 0×0.002 + 0.1×0.096 + 0.1×0.024
= 1.02 kW
Heat dissipated inside the enclosure (→ E2)
[35] P_DC = ∑ DC P_i·d_i·k_i
= 120 W
= 120 W
DC demand
[36] P_PSU_req = margin * P_DC / f_derate
= 1.25 × 120 W / 1
= 150 W
Minimum PSU rating
[37] S_T_req = margin * S_dem
= 1.25 × 12.993 kVA
= 16.241 kVA
Minimum transformer rating for the running load
[38] S_peak = S_dem + max(m_i·P_i/cos∅_i - S_i)
= 12.993 + 70.349 (Duty pump starting)
= 83.342 kVA
Peak apparent power while the largest load starts
[39] S_T_req_start = S_peak / k_T
= 83.342 kVA / 3
= 27.781 kVA
Minimum transformer rating for the starting peak
[40] S_T_min = max(S_T_req, S_T_req_start)
= max(16.241 kVA, 27.781 kVA)
= 27.781 kVA
Minimum transformer rating (running load and starting peak)
[41] I_b = ∑ backed P_i·d_i·k_i/(U_b·∅_conv)
= 120 / (24 × 1)
= 5 A
Battery discharge current

[42] Ah = I_b * t_auto
= 5 A × 2 h
= 10 A·h
Discharge capacity
[43] C_req = Ah / (DoD * eta_b * k_temp * k_age)
= 10 A·h / (0.8 × 0.95 × 1 × 0.8)
= 16.447 A·h
Minimum battery capacity
[45] u_PSU = P_PSU_req / P_PSU
= 150 W / 240 W
= 0.625
PSU utilisation
[47] u_T = S_T_req / S_T
= 16.241 kVA / 25 kVA
= 0.64964
Transformer utilisation
[48] u_inrush = S_peak / (k_T * S_T)
= 83.342 kVA / (3 × 25 kVA)
= 1.1112
Inrush utilisation
[50] u_batt = C_req / C_batt
= 16.447 A·h / 17 A·h
= 0.96749
Battery utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
P_conn	22.2 kW	Connected load
P_dem	11.182 kW	Demand active power
S_dem	12.993 kVA	Demand apparent power
I_3ph	18.462 A	Three-phase bus current
I_1ph	0.87872 A	Single-phase bus current
I_DC	5 A	DC bus current
Phi	1.02 kW	Heat dissipated inside the enclosure
P_PSU_req	150 W	Minimum PSU rating
S_T_req	16.241 kVA	Minimum transformer rating, running load
S_peak	83.342 kVA	Peak apparent power at start
S_T_req_start	27.781 kVA	Minimum transformer rating, starting peak
S_T_min	27.781 kVA	Minimum transformer rating, running and starting
I_b	5 A	Battery discharge current
Ah	10 A·h	Discharge capacity
C_req	16.447 A·h	Minimum battery capacity
u_PSU	0.625	PSU utilisation
u_T	0.64964	Transformer utilisation
u_inrush	1.1112	Inrush utilisation

SYMBOL	VALUE	DESCRIPTION
u_batt	0.96749	Battery utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Starting peak against transformer overload	u_inrush = 1.1112	u_inrush ≤ 1	1.11	
PASS	Battery capacity against required	u_batt = 0.96749	u_batt ≤ 1	0.967	
PASS	Transformer rating against required	u_T = 0.64964	u_T ≤ 1	0.65	
PASS	PSU rating against required	u_PSU = 0.625	u_PSU ≤ 1	0.625	

7. Applicability

WARNING — record: Tabulated values not taken from a reference record: f_derate (PSU derating); DoD, eta_b, k_temp, k_age (Battery factors). Enter each as one so its source, edition and licence travel with it.
WARNING — input.P_add: P_add (Additional three-phase load (linked from E1's RMS power)) was not entered; the template default 0 kW is used.
WARNING — input.cosphi_add: cosphi_add (Power factor of the additional load) was not entered; the template default 0.85 is used.
WARNING — input.eta_add: eta_add (Efficiency of the additional load) was not entered; the template default 0.9 is used.
WARNING — input.inrush_add: inrush_add (Starting multiple of the additional load) was not entered; the template default 1 is used.
WARNING — input.heat_add: heat_add (Share of the additional load's input power released inside the enclosure (a linked machine is outside it: 0)) was not entered; the template default 0 is used.
WARNING — derate: f_derate: the ambient is above 40 °C but no PSU derating is entered; read the factor from the manufacturer's curve.

8. Verification

Template elec.load-budget@1.2.0 Content hash dc7b6b8e4454e8eb Engine calc-engine/1.0.0
Known-answer cases passed 4/4 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0